Autonomous Code Review & Refactoring Agent

Practical Assessment Report - LangGraph, FastAPI, Docker

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GitHub Repository: https://github.com/kavya1407/Autonomous-Code-Review-Agent

Assessment Topic: Topic 2 - Autonomous Code Intelligence: Build an AI-Powered Agent

Tech Stack: Python, LangGraph, FastAPI, GitHub REST API, Docker, pytest, flake8, black

Test Results: 17 / 17 Automated Tests Passing (100%)

1. Executive Summary & Business Context

Enterprise development teams require automated systems to review incoming Pull Requests (PRs), execute linters inside sandboxed environments, propose refactoring templates, and require human engineer confirmations before mutating repositories. This project implements an end-to-end autonomous agent using LangGraph, featuring stateful transitions, persistent checkpointers, Human-in-the-Loop (HITL) interrupt gates, and a secure subprocess linter sandbox.

2. LangGraph Stateful Pipeline Architecture

The review pipeline is constructed as a LangGraph StateGraph tracking 'pr_url', 'code_contents', 'lint_errors', 'refactored_code', 'is_approved', 'reviewer_feedback', and chronological audit 'history'.

Workflow Nodes:
- fetch_pr_node: Pulls modified files from GitHub PR via REST API.
- lint_code_node: Runs flake8 and black --check inside a temporary filesystem sandbox.
- propose_refactor_node: Invokes LLM (or deterministic engine) to resolve errors and apply feedback.
- apply_changes_node: Commits approved code back to the branch via GitHub API.

HITL Interrupt Gate: Execution pauses before repository mutation using interrupt_before=['apply_changes_node']. When feedback is provided instead of approval, the state machine rolls back to propose_refactor_node, regenerates the proposal incorporating reviewer directives, and repauses at the write gate.

3. Custom Python Execution Sandbox Tool

Implemented via LangChain's @tool decorator in 'run_linter_tool'. The tool creates an isolated temporary directory via tempfile.TemporaryDirectory, writes source code, and spawns subprocesses for flake8 and black with strict 15-second execution timeouts. It isolates execution from the host and safely captures stdout and stderr.

4. FastAPI Endpoints & Integration

POST /api/webhooks/github	Ingests PR events, extracts parameters, and triggers stateful thread.
POST /api/reviews/{thread_id}/approve	Accepts approval/rejection with optional feedback to resume or rollback thread.
GET /api/reviews/{thread_id}/state	Inspects active checkpoint state, next nodes, and history log.
GET /health	System health and persistence configuration status.

5. Verification & Validation Results

All three required project verification workflows were executed and validated:

1. Test Case 1 (Webhook Ingestion & HITL Gate): PR webhook received, linter executed inside sandbox, and execution successfully halted before apply_changes_node with status 'paused_for_approval'.

2. Test Case 2 (Feedback & Proposal Re-route): Reviewer rejected with 'Use list comprehensions instead of loops'. The graph re-routed to propose_refactor_node, regenerated draft with sum([item for item in items]), and repaused before the write gate.

3. Test Case 3 (Approval & Commit Write Back): Reviewer approved with approved=true. Graph resumed, executed apply_changes_node, committed changes to GitHub branch, and completed thread with status 'completed'.

6. Automated Test Suite (17 / 17 Passing)

- tests/test_sandbox_tool.py: Verified flake8 error detection (F401, E401), syntax errors (E999), and clean code.
- tests/test_graph_pipeline.py: Verified LangGraph interrupt gates, rejection rollback loops, and completion.
- tests/test_api_endpoints.py: Verified Test Cases 1, 2, 3 end-to-end via FastAPI TestClient.
- tests/test_routing_and_rollback.py: Verified multi-turn review feedback and checkpointer history snapshots.
- tests/test_github_service.py: Verified PR URL parsing and mock commit operations.
- tests/test_refactor_service.py: Verified markdown extraction and feedback instruction handling.

7. Submission Checklist

[x] GitHub Public Repository Link	https://github.com/kavya1407/Autonomous-Code-Review-Agent
[x] OpenAPI Specification	openapi.json included in repository root
[x] README Documentation	Full architecture diagram, setup, and test verification guide
[x] Docker Infrastructure	Dockerfile and docker-compose.yml configured
[x] Git Commits	10 progressive commits demonstrating incremental milestones